

Study on ionic transport mechanism and interactions between salt and polymer chain in PAN based solid polymer electrolytes containing LiCF_3SO_3

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Abstract

Solvent-free polymer electrolytes based on polyacrylonitrile (PAN) including lithium trifluoromethane sulfonate (LiCF_3SO_3) were prepared. All measurements were performed on samples with various salt contents between 0 and 70 wt.%. Interactions between ion and polymer or interion were investigated by using Fourier-transform infrared (FT-IR) spectroscopy. X-ray diffraction measurement was also used for the analysis of aggregation states of polymer matrix and salt complex. By analysis of $\nu_s(\text{SO}_3)$ mode, it was confirmed that larger ion clusters were formed quickly above salt concentration of 50 wt.% and ionic conductivity increased dramatically in this composition region.
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1. Introduction

A polymer electrolyte with high ionic conductivity and good mechanical properties is one of the goals that researchers in this field have been pursuing for decades [1]. Common features of so-called traditional polymer electrolytes are that the cation coordinates with polymer chain, and ionic transport occurs in amorphous regions of materials, associated with polymer segmental motion [2].

Solid polymer electrolyte (SPE) based on PAN with good mechanical properties has had many advantages in the application of secondary rechargeable battery. First, the ionic conductivity of SPE based on PAN depends on not segmental motion of polymer but ionic mobility. That is, these systems show decoupled ion transport behavior in wide range of temperature [3]. Second, PAN based polymer electrolyte

has a very unique mechanism in polymer-in-salt system with much higher salt content than in traditional salt-in-polymer electrolyte. Generally, when the salt is added more than its critical concentration in traditional amorphous polymer electrolyte, ion clusters are formed in polymer electrolyte, and ionic mobility decreases. A decrease of ion mobility leads to the decrease of ionic conductivity. However, ion clusters in polymer-in-salt system make percolation path, and cation is easy to move in the entire electrolyte through the percolation path. In relation to that, Ferry et al. [2] suggest that a more efficient ionic transport mechanism appears to be associated with a higher degree of ionic aggregation, the dramatic enhancement of the ionic conductivity reflects dynamic connectivity effects in a phase-separated electrolyte passing through a “smeared” percolation threshold.

According to Bushkova et al. [4] convinced that when a critical cluster concentration has been achieved, all separate single clusters come into contact with each other thus forming an infinite cluster, promoting fast cationic transport process in the entire electrolyte.

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In this study, we prepared PAN based solvent-free electrolytes containing LiCF_3SO_3 and investigated interactions between ion and polymer segment or interion by using Fourier transform infrared spectroscopy (FT-IR) analysis and X-ray diffractometer.

2. Experimental

2.1. Materials

Salts, lithium trifluoromethanesulfonate (LiCF_3SO_3 , Aldrich Chemical Co.) and magnesium trifluoromethanesulfonate ($\text{Mg}(\text{CF}_3\text{SO}_3)_2$, Aldrich Chemical Co.) were dried at 160°C for 12 h. And then they were dried under vacuum at room temperature prior to their use. Polyacrylonitrile (PAN, $M_w = 150,000$, Aldrich Chemical Co.) and Dimethylformamide (DMF, J.T. Baker) were used without purification process.

2.2. Preparation of polymer electrolytes

Five grams of PAN powder was added to the 10 ml of DMF solvent, and the mixture was stirred at 60°C until the polymer solution was homogeneous. LiCF_3SO_3 was introduced variously to polymer solution from 10 to 70 wt.%, and it was stirred for 6 h at 60°C . The PAN/salt films were prepared by casting the solution onto glass plates. The films were allowed to dry in vacuum oven ($<10^{-2}$ Torr) for 24 h at 40°C . Further drying was achieved for 24 h at 80°C to remove the residual solvent. The polymer electrolyte films were prepared by casting the solution directly on stainless steel, and dried in a vacuum oven for 48 h. These films were prepared by using a press at temperature between 160 and 170°C under 5 or 6 metric tons.

2.3. Characterization

FT-IR spectra were measured at ambient temperature by using a Jasco 460 Plus spectrometer in the range between 4000 and 400 cm^{-1} . The XRD equipment used in this work was Rigaku model D/Max-2400 ($\lambda = 1.5405\text{ \AA}$, scanning rate of 2°C/min). Impedance spectra of SPEs based on PAN were recorded in the frequency range from 1 MHz to 10 Hz by an electrochemical impedance analyzer (Zahner Elektrik IM6) at room temperature. All cell construction activities were carried out in an argon atmosphere with O_2 and H_2O concentrations less than 1 ppm.

3. Results and discussion

3.1. FT-IR spectroscopic analysis

$\text{C}\equiv\text{N}$ group stretching spectra of SPEs based on PAN is presented in Fig. 1. A new component (at 2270 cm^{-1} in Li-

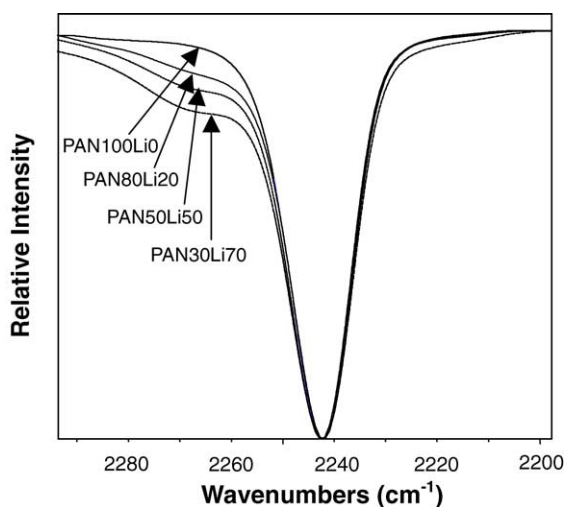


Fig. 1. $\text{C}\equiv\text{N}$ stretching modes of SPEs based PAN.

based PAN) split from the $\text{C}\equiv\text{N}$ stretching band of free $\text{C}\equiv\text{N}$ at 2241 cm^{-1} was observed. It represents the Li^+ ion coordination with the $\text{C}\equiv\text{N}$ group of PAN [5]. The relative peak intensity at 2270 cm^{-1} increases with salt concentration due to an increase in the number of carrier ion.

$\delta_s(\text{CF}_3)$ and $\nu_s(\text{SO}_3)$ internal modes of CF_3SO_3^- anion are particularly sensitive to the change in the local anionic environment [2,5–7]. The corresponding bands, which obtained from $\delta_s(\text{CF}_3)$ and $\nu_s(\text{SO}_3)$ mode of triflate anion, reflect the dissolution of the triflate ionic species in Li-based PAN. The relative intensities of $\delta_s(\text{CF}_3)$ and $\nu_s(\text{SO}_3)$ mode provide quantitative information regarding various ionic species distribution.

$\delta_s(\text{CF}_3)$ peak of SPEs based on PAN is shown in Fig. 2. The 760 cm^{-1} component in Li-based PAN electrolyte shows $\delta_s(\text{CF}_3)$ peak. The $\delta_s(\text{CF}_3)$ peaks in Li-based PAN electrolyte were shifted from 760 to 767 cm^{-1} with salt concentration. It indicates that ion clusters are formed in entire polymer electrolyte with increasing salt concentration due to strong

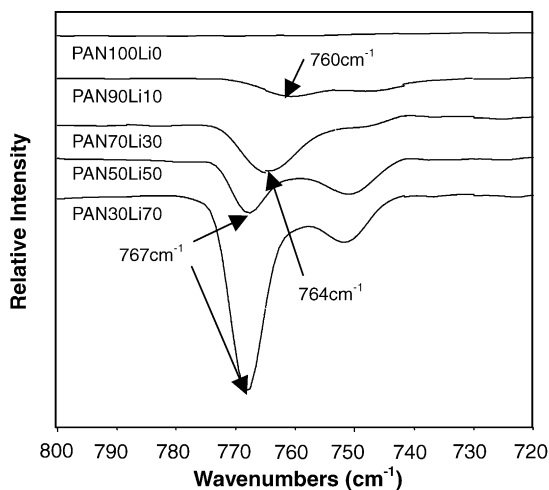


Fig. 2. $\delta_s(\text{CF}_3)$ modes of SPEs based on PAN.

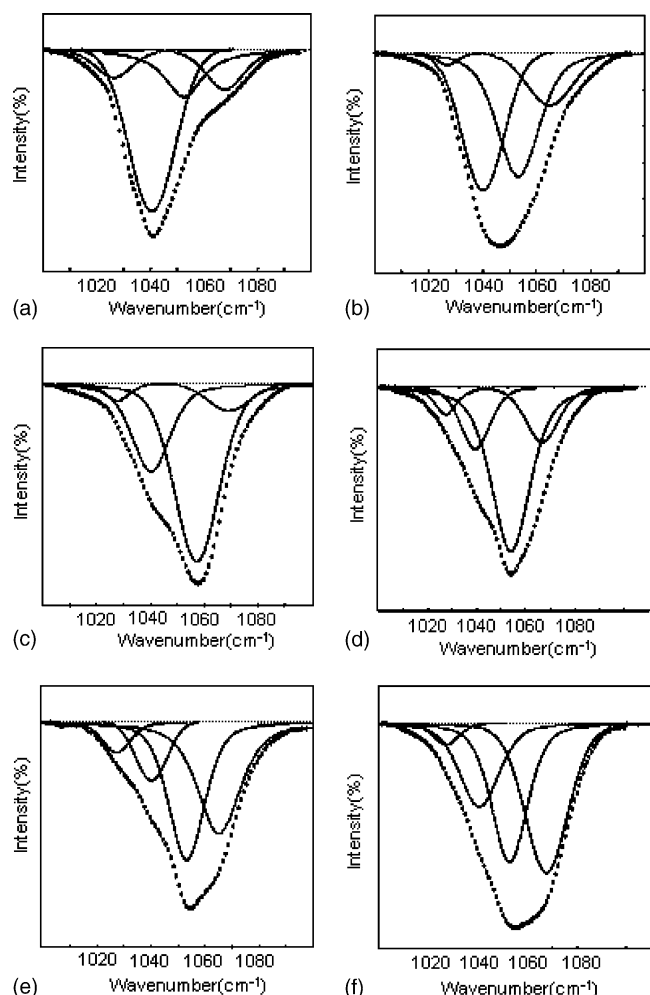


Fig. 3. $\nu_s(\text{SO}_3)$ modes of Li-based PAN electrolytes: (a) PAN80Li20, (b) PAN70Li30, (c) PAN60Li40, (d) PAN50Li50, (e) PAN40Li60, and (f) PAN30Li70.

interaction of ions. These interactions of ions lead to a change of $\delta_s(\text{CF}_3)$ mode, a peak of 760 cm^{-1} component shifts to 767 cm^{-1} , and its intensity increases.

In Fig. 3, FT-IR spectra of $\nu_s(\text{SO}_3)$ spectral region of triflate ion by curve deconvolution are presented. According to many studies [8,9], it could be assigned that the components at 1027 , 1040 , 1053 , and 1065 cm^{-1} are ascribed to free ion, ion pairs, ion triple, and larger ion clusters, respectively. Intensity of ion triple peak (1053 cm^{-1}) increased quickly with increasing salt content below 50 wt.%. It indicates that ion clusters are formed in entire polymer electrolyte with increasing salt content because interaction of ions is stronger than that between ion and polymer. When the salt content further increased (above 50 wt.%), it observed that the 1065 cm^{-1} components increased dramatically. It is believed that when the salt is added above the critical concentration, large ion clusters are formed due to strong interaction of ions and these ion clusters come into contact with each other, and then efficient percolation paths are constructed.

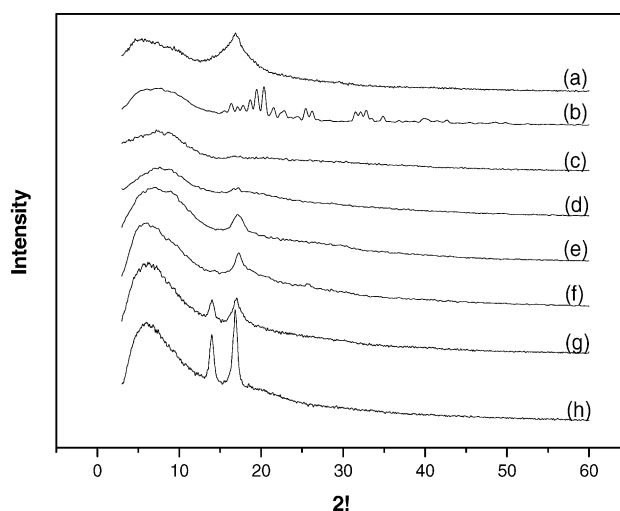


Fig. 4. The XRD patterns of: (a) pure PAN, (b) pure LiCF_3SO_3 , (c) PAN80Li20, (d) PAN70Li30, (e) PAN60Li40, (g) PAN40Li60, and (h) PAN30Li70.

3.2. X-ray diffraction studies

Fig. 4 shows the XRD patterns of Li-based PAN with various salt concentration compared with pure LiCF_3SO_3 and PAN. The XRD pattern of pure PAN shows a crystalline peak at $2\theta = 17^\circ$ and it corresponds to orthorhombic PAN (1 1 0) reflection [10,11,12]. The peak of $2\theta = 17^\circ$ disappears at salt content of 20 wt.% (Fig. 4c). It can be seen that salt is dissociated well within polymer matrix, and the PAN-based lithium salt complexes exhibit the amorphous state [12]. However, when salt is added above 20 wt.%, the peak at $2\theta = 17^\circ$ appears and sharpens gradually. As well, a new sharp component at $2\theta = 14^\circ$ appears above salt concentration of 60 wt.%. This peak may arise from that large ion clusters are formed by the strong interaction of ions, and the ordering of large ion clusters, concerning with ion percolation paths, is

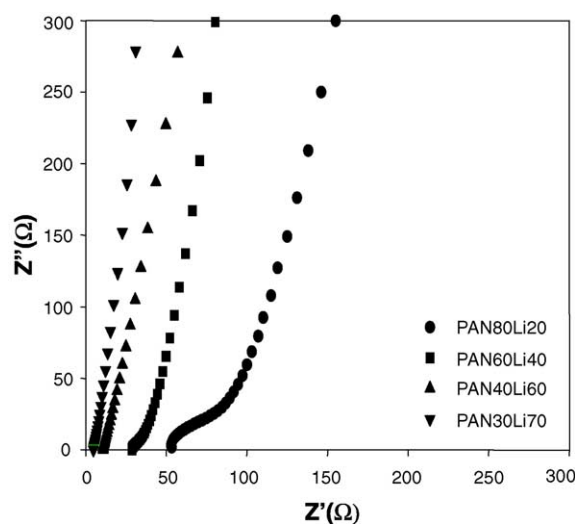


Fig. 5. Cole-Cole plots obtained from ac impedance analysis of Li-based PAN.

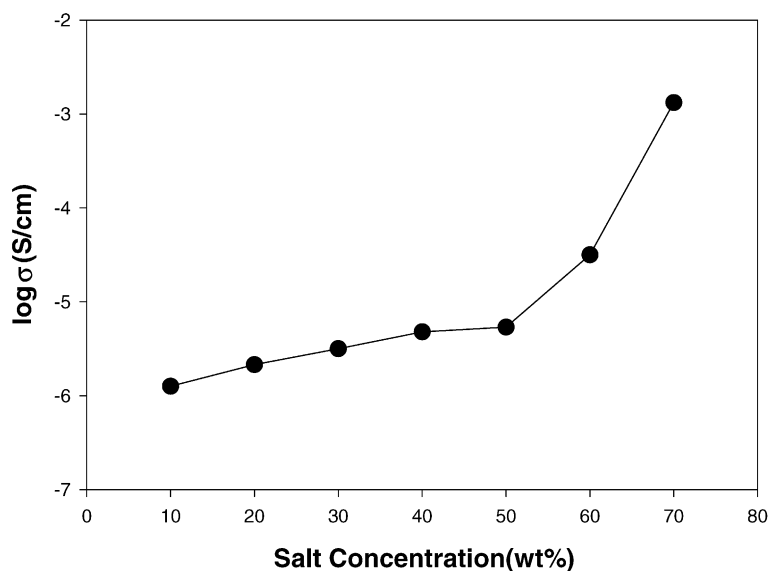


Fig. 6. Ionic conductivity of Li-based PAN with various salt content.

accelerated with salt concentration. And, the ordering of polymer matrix was also accelerated.

3.3. Impedance plot

Cole–Cole plots obtained from ac impedance analysis at room temperature are shown in Fig. 5. A semi-circle corresponding to the interfacial impedance did not observe above salt content of 60 wt.%. And as salt content increases, a semi-circle disappears gradually. Ionic conductivity of Li-based PAN dramatically increased because interfacial impedance disappeared and bulk impedance largely decreased by the change of salt content from 50 to 70 wt.%.

3.4. Ionic conductivity

Fig. 6 shows ionic conductivities of Li-based PAN with various salt concentrations. Ionic conductivity increases gradually until 50 wt.% of salt concentration, however ionic conductivity abruptly increases between 50 and 70 wt.%. These results correspond to previous $\nu_s(\text{SO}_3)$ mode analysis well. When the salt content is extremely high (close to the limit of the solubilization of the polymer), efficient percolation paths are formed and contribute the most to the ionic conductivity of the electrolyte.

Consequently, the highest ionic conductivity was achieved at 70 wt.% salt content, and its value was 1.32×10^{-3} S/cm.

4. Conclusions

Solvent-free polymer electrolytes based on polyacrylonitrile (PAN) including lithium trifluoromethane sulfonate (LiCF_3SO_3) with various salt contents were prepared. By an-

alyzing the Fourier-transform infrared (FT-IR) spectrum of the solid polymer electrolytes, it confirmed that large ion clusters are formed in entire polymer electrolyte with the increase of salt content and efficient ion percolation paths are constructed above the critical salt concentration. XRD measurements also indicated that strong interactions were occurred between ionic species and polymer. Salt in low concentration plays a role as a plasticizer, but on the other hand it in high content makes efficient percolation paths to move cation easily. Ionic conductivity of the Li-based PAN increases dramatically by nearly two orders of magnitude between 50 and 70 wt.%. The highest ionic conductivity of 1.32×10^{-3} S/cm was achieved at 70 wt.% salt content.

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